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B.Tech. DEGREE EXAMINATION, DECEMBER 2022
Third Semester

18BTC102J – CELL BIOLOGY

(For the candidates admitted from the academic year 2020-2021 to 2021-2022)

Note:

- (i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
- (ii) **Part - B** should be answered in answer booklet.

Time: 2½ Hours

Max. Marks: 75

PART – B (5 × 10 = 50 Marks)

Answer **ALL** Questions

Marks BL CO PO

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| 26. a. Discuss in detail about various molecular composition of cells. | 10 | 3 | 1 | 1 |
| (OR) | | | | |
| b. Describe the structure of cell membrane and mention its function with appropriate diagram. | 10 | 3 | 1 | 1 |
| 27. a. Describe how lipid and polysaccharide metabolism occur in Golgi. Add a note on protein sorting. | 10 | 4 | 2 | 3 |
| (OR) | | | | |
| b. Write a detailed note on the structure and functions of mitochondria. | 10 | 3 | 2 | 3 |
| 28. a. Discuss about the role of actin and myosin in muscle contraction. | 10 | 4 | 3 | 4 |
| (OR) | | | | |
| b. What are the various types of cell-cell interactions? | 10 | 3 | 3 | 4 |
| 29. a. Mention the importance of cell signaling and the pathway of intracellular signal transduction. | 10 | 4 | 4 | 4 |
| (OR) | | | | |
| b. Write a detailed essay on the stem cells and their usages in therapy. | 10 | 3 | 4 | 4 |
| 30. a. Define Cancer. Discuss in detail about oral cancer along with treatment. | 10 | 4 | 5 | 4 |
| (OR) | | | | |
| b. Discuss in detail about neurodegenerative diseases with a note on Tau hypothesis. | 10 | 4 | 6 | 5 |

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